Eduardo Alvarez

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EDUCATION

University of Nebraska-Lincoln — Lincoln, NE

Bachelor of Science in Mechanical Engineering — Expected Graduation May 2028

Relevant Coursework: Statics, Dynamics, Mechanics of Materials, Thermodynamics, Machine Design

ENGINEERING EXPERIENCE

Husker Motorsports Formula SAE Team — Lincoln, NE

Suspension/Vehicle Dynamics Director — August 2024-Present

- **Led vehicle dynamics development** by analyzing tire behavior, load transfer, suspension kinematics, and handling balance, resulting in a suspension design and setup decisions
- **Analyzed** Formula SAE Tire Test Consortium data in self-made MATLAB script to compare relationships of lateral force, slip angle, cornering stiffness, pressure, inclination angle, and vertical load, resulting in data driven decision for the best tire and setup.
- **Directed the design and integration** of the inboard and outboard suspension systems, including rockers, pushrods, control arms, uprights, hubs, and mounting components, resulting in a fully package suspension system with verified clearance.
- **Designed** a decoupled inboard suspension actuation system in SolidWorks; performed component level FEA to verify stress met required factors of safety and executed clearance checks across inboard packaging.
- **Operated** CNC mills and lathes, plus manual mill/lathe equipment, to manufacture high-quality components to print specifications and tight tolerance requirements.

PROJECTS

Electrathon Electric Vehicle Team — Olathe, KS

Olathe Northwest High School Engineering Project — August 2022-May 2024

- **Designed**, **installed**, and **integrated** electrical systems for a student-built electric endurance vehicle, including battery wiring, motor controllers, sensors, and power distribution.
- **Troubleshot** electrical components and battery systems to support reliable vehicle performance during competition.

ADDITIONAL EXPERIENCE

Husker Hub, University of Nebraska–Lincoln — Lincoln, NE

Peer Specialist — August 2024-Present

- Guided students and families on FAFSA, scholarships, grants, loans, enrollment verification, and billing; clearly explained processes to ensure accurate, compliant financial aid information.
- Assisted 1,000+ students and families weekly with financial aid questions by providing clear, accurate guidance to support informed funding decisions.

Rigo's Landscape LLC — Olathe, KS

Landscaper — June 2013-August 2023

- Collaborated with cross-functional crews to deliver landscaping projects (patios, retaining walls, irrigation) valued up to $100,000, meeting schedule and client specifications.
- Diagnosed and repaired mechanical issues on heavy and small equipment, applying hands-on engineering principles to maintain uptime and operational efficiency.

TECHNICAL SKILLS

CAD:	SolidWorks (3D modeling, assemblies, technical drawings), Fusion 360 (3D modeling, assemblies, CAM), AutoCAD (2D Drafting, Layouts),
Engineering Analysis:	MATLAB, Finite Element Analysis (FEA), Vehicle Dynamics
Manufacturing:	CNC Milling, CNC Turning, Manual Mills & Lathes, Technical Drawings